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Chemistry Paper 3A WASSCE (SC), 2016

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Weakness/
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Question 3A

3. (a) What difference in physical properties enable the separation of mixtures by:

- | | |
|-------|--------------------------|
| (i) | simple distillation; |
| (ii) | paper chromatography; |
| (iii) | fractional distillation. |

[3 marks]

(b) Give a reason for each of the following practices during titration in the laboratory.

- | | |
|------|---|
| (i) | White tile is placed under the conical flask. |
| (ii) | Burette readings are always recorded to two decimal places. |

[2 marks]

(c) Calculate the volume of 2.5 moldm⁻³ stock HCl required to prepare 500 cm³ of 0.20 moldm⁻³ HCl.

The question tested candidates' familiarity with the practical activities suggested in their teaching syllabuses.

Observation

Majority of the candidates did not have knowledge of paper chromatography. They could not state a reason each for placing white tile under conical flask. and recording burette reading to 2 decimal places during titration. However, they showed understanding of dilution in stoichiometry and chemical reactions.

Expected answers include:

Question 3

- | | | |
|-----|------|--|
| (a) | (i) | Wide difference in boiling points of components (for miscible liquids) / Difference in physical states (for liquid – solid solution) |
| | (ii) | Difference in adsorption / retention rate of components |

(iii) Close difference in boiling points of components

(b) (i) To make the end point colour change easily visible / in order not to overshoot the end point

(ii) To ensure (high) accuracy of the titre value

(c) Using the dilution formula

$$C_{\text{conc}} \times V_{\text{conc}} = C_{\text{DIL}} \times V_{\text{DIL}} \text{ (or } C_1 V_1 = C_2 V_2 \text{)}$$

$$V_{\text{conc}} = \frac{C_{\text{DIL}} \times V_{\text{DIL}}}{C_{\text{conc}}}$$

$$= \frac{0.200 \times 500}{2.5} = 40.0 \text{ cm}^3$$

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